

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I Candana Ramdani (192021063) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

C. Ramdani

20/20
ADH

Remote Rural Branch office - Internet connection Networks and Security

Introduction:

- The multinational company is opening a branch office in a remote rural area where high speed wired internet is unavailable.
- The office requires internet connectivity for cloud application, video conferencing and secure application, video conferencing and headquarter.
- Secure budget is limited. The solution must provide balance between cost, reliability, bandwidth, security and future scalability.

A) Constraints:

1. high speed wired internet is unavailable
2. the office requires cloud based.

application

3) video conferencing requires high bandwidth and low latency

4) The connectivity budget is limited.

B. Recommended Internet connection

Recommendation is for wireless broadband connection and primary connection depending on cellular availability

Satellite Internet connection

Proposed arrangement:

Primary: High speed Fixed Wireless Internet

Backup: Satellite Internet

Application Requirement

High density areas

more coverage capacity should be

x lecture hall

x libraries

x laboratories

x Hostel

x Student common area

D. Frequency Band Selection

Advantage:

x large range

x Better wall penetration

Disadvantage:

x more interference

x fewer non-overlapping channels

Suit 2

Advantage :

* more available channels

* higher throughput

* lower interference compared with 2.4GHz

Q. Network Segmentation

VLAN

User

VLAN 10

Student

VLAN 20

Faculty

VLAN 30

Administration

VLAN 40

Guest user

VLAN 50

Network management

Conclusion:

The recommended design is centrally managed enterprise wireless network.

University Wireless Network for more than 3,000 Students

Introduction:

< The university had campus-wide wireless connectivity for more than 3,000 students across classrooms, laboratories, and hostels.

< The network must support many simultaneous users while minimizing interference.

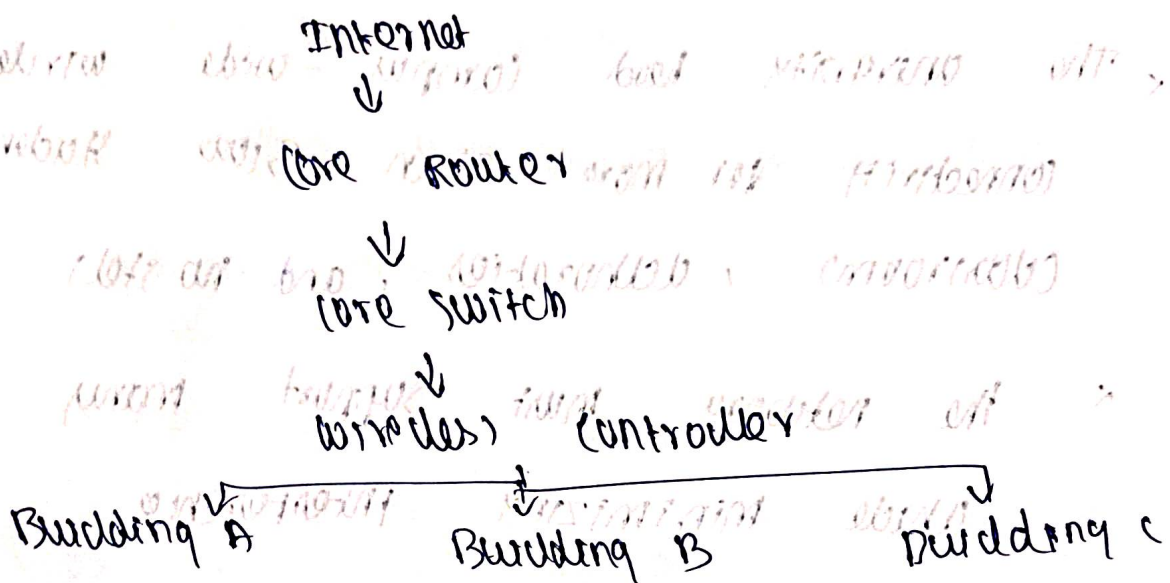
A (on front)

1. more than 3,000 students
2. large number of simultaneous connections
3. campus wide coverage
4. (class) rooms, laboratories, and hostels

different usage patterns

4) Secure authentication i) required

B. proposed network design



C) Access point placement:

Access point should not simply be placed at equal distance

- * student density
- * Building structure

Excerpted

- * wall thickness

Secure Network Architecture for Financial Institution

Introduction:

* A Financial institution handles high sensitive customer and banking information

* The organization must maintain good network performance and comply with strict security policies

* A constraint =

1. Sensitive customer information must be protected

2. Employee required access to authorized access

3. Online banking services must remain available.

9) Unauthorized access must be detected and prevented

5) the architecture must satisfy

B. proposed secure Architecture:

